20.2: Biologists use DNA technology to study gene expression and function

To see how a biological system works, scientists seek to understand the functioning of the system's component parts. Analysis of when and where a gene or group of genes is expressed can provide important clues about their function and how they contribute to the organism as a whole.

Analyzing Gene Expression

Biologists driven to understand the assorted cell types of a multicellular organism, cancer cells, or the developing tissues of an embryo first try to discover which genes are expressed by the cells of interest. The most straightforward way to do this is usually to identify the mRNAs being made. We'll first examine techniques that look for patterns of expression of specific individual genes. Next, we'll explore ways to characterize groups of genes being expressed by cells or tissues of interest. As you will see, all of these procedures depend in some way on base pairing between complementary nucleotide sequences.

Studying the Expression of Single Genes

Suppose we have cloned a gene that we suspect plays an important role in the embryonic development of *Drosophila melanogaster* (the fruit fly). The first thing we might wonder is which embryonic cells express the gene—in other words, where in the embryo is the corresponding mRNA found? We can detect the mRNA in an embryonic sample using nucleic acid hybridization with molecules of complementary sequence to the mRNA we want to follow. Using our cloned gene as a template, we can synthesize a short, single-stranded nucleic acid (either RNA or DNA) complementary to the mRNA of interest; this is called a **nucleic acid probe**. For example, if part of the sequence on the mRNA were

Five prime. C, U, C, A, U, C, A, C, C, G, G, C. Three prime.

then we would synthesize this single-stranded DNA probe:

Three prime. G, A, G, T, A, G, T, G, G, C, C, G. Five prime. The strand is highlighted yellow.

Each probe molecule is labeled during synthesis with a fluorescent tag so we can follow it. A solution containing probe molecules is applied to *Drosophila* embryos, allowing the probe to hybridize specifically with any complementary sequences on the many mRNAs in embryonic cells in which the gene is being transcribed. Because this technique allows us to see the mRNA in place (or *in situ*, in Latin) in the intact

organism, this technique is called ***in situ*hybridization**. Different probes can be labeled with different fluorescent dyes, sometimes with strikingly beautiful results **(Figure 20.9)**.

Figure 20.9

```
    Determining where single genes are expressed by *in situ*
hybridization analysis.
```

The yellow probe hybridizes with mRNAs in cells that are expressing the wingless (w g) gene. The blue probe hybridizes with mRNAs that are expressing the engrailed (e n) gene.

This Drosophila embryo was incubated in a solution containing DNA probes for five different mRNAs, each probe labeled with a different fluorescently colored tag. The embryo was then viewed using fluorescence microscopy. Each color marks where a specific gene is expressed as mRNA. The arrows from the groups of yellow and blue cells above the micrograph show a magnified view of nucleic acid hybridization of the appropriately colored probe to the mRNA. The thorax (trunk) and abdomen are made up of repeating segments. Yellow cells (expressing the wg gene) interact with blue cells (expressing the en gene); their interaction helps establish the pattern in a body segment.

The diagram is as follows.

•The yellow DNA probe hybridizes with mRNAs in cells that are expressing the wingless (wg) gene, which encodes a secreted signaling protein. Sequences as follows: Top strand (yellow probe): 3 prime T A A C G G T T C C A G G 5 prime). Bottom strand (5 prime A T T G C C A A G G T C G 3 prime). Cells expressing the w g gene are yellow.

•The blue DNA probe hybridizes with mRNAs that are expressing the engrailed (en) gene, which encodes a transcription factor. Sequences shown are: Top strand (blue probe): 3 prime C T C A A G T T G C T C T 5 prime). Bottom strand: 5 prime G A G T T C A A C G A G A 3 prime. Cells expressing the e n gene are blue.

•Fluorescent micrograph (scale: 50 micrometers) of the embryo shows clear yellow regions at both ends of the head segment, and at the posterior boundaries of each segment T1, T2, T3, A1, A2, A3, A4 and A5, where T indicates thorax segments from anterior to posterior and A indicates abdomen segments from anterior to posterior. Blue regions are just adjacent to yellow in the thorax and abdomen segments.

Other mRNA detection techniques may be preferable for comparing the amounts of a specific mRNA in several samples at the same time—for example, in different cell types or in embryos of different stages. One method that is widely used is called the **reverse transcriptase polymerase chain reaction**, or **RT-PCR**.

RT-PCR begins by turning sample sets of mRNAs into double-stranded DNAs with the corresponding sequences. First, the enzyme reverse transcriptase (from a retrovirus; see Figure 19.9) is used to synthesize a complementary DNA copy (a *reverse transcript*) of each mRNA in the sample **(Figure 20.10)**. The mRNA is then degraded by addition of a specific enzyme, and a second DNA strand, complementary to the first, is synthesized by DNA polymerase. The resulting double-stranded DNA is called **complementary DNA (cDNA)**. (Made from mRNA, cDNA lacks introns and can be used for protein expression in bacteria, as mentioned earlier.) To analyze the timing of expression of the *Drosophila* gene of interest, for example, we would first isolate all the mRNAs from different stages of *Drosophila* embryos and make cDNA from each stage **(Figure 20.11)**.

Figure 20.10

```
Making complementary DNA (cDNA) from eukaryotic genes.
```

Steps 1-5 diagrammed in the synthesis of cDNA using reverse transcriptase and isolated mRNA. The cDNA carries the complete coding sequence of the gene, but no introns.

Complementary DNA is made in a test tube using mRNA as a template for the first strand. Only one mRNA is shown after step 1, but the final collection of cDNAs would reflect all the mRNAs present in the cell.

A 5-step process is shown for making complementary DNA (cDNA) from eukaryotic genes.

1.Reverse transcriptase is added to a test tube containing mRNA isolated from a sample of cells (DNA in nucleus and mRNAs in cytoplasm).

2. Reverse transcriptase makes the first DNA strand using the mRNA as a template and a short poly-dT as a DNA primer. Diagram shows mRNA: 5 prime (unlabeled span of bases), poly A tail 3 prime. DNA primer synthesized with reverse transcriptase from 5 prime to 3 prime from mRNA template.
3. mRNA is degraded by another enzyme. Diagram shows mRNA molecule breaking apart.
4. DNA polymerase synthesizes the second DNA strand, using a primer in the reaction mixture. (Several options exist for primers).
5. The result is cDNA, which carries the complete coding sequence of the gene but no introns.

Figure 20.11

Research Method: RT-PCR Analysis of the Expression of Single Genes

Application

RT-PCR uses the enzyme reverse transcriptase (RT) in combination with PCR and gel electrophoresis. RT-PCR can be used to compare gene expression between samples—for instance, in different embryonic stages, in different tissues, or in the same type of cell under different conditions.

Technique

In this example, samples containing mRNAs from six embryonic stages of *Drosophila* were analyzed for a specific mRNA as shown below. (In steps 1 and 2, the mRNA from only one stage is shown.)

Complementary D N As are synthesized from m R N A and then amplified using P C R. Gel electrophoresis reveals amplified D N A products only in samples containing m R N A transcribed from the specific Drosophila gene.

Step 1: c D N A synthesis is carried out by incubating the m R N As with reverse transcriptase and other necessary components. Step 2: P C R amplification of the sample is performed using primers specific to the Drosophila gene of interest. Step 3: Gel electrophoresis will reveal amplified D N A products only in samples that contained m R N A transcribed from the specific Drosophila gene. Gel electrophoresis shows embryonic stages 1-6, with bands in the middle of the gel at 2-6. Results: The m R N A for this gene first is expressed at embryonic stage 2 and continues to be expressed through embryonic stage 6. The size of the amplified fragment depends on the distance between the primers that were used, not on the size of the m R N A.

Results

The mRNA for this gene is expressed from stage 2 through stage 6. The size of the amplified fragment (shown by its position on the gel) depends on the distance between the primers that were used (not on the size of the mRNA).

Next in RT-PCR is the PCR step (see Figure 20.7). As you will recall, PCR is a way of rapidly making many copies of one specific stretch of double-stranded DNA, using primers that hybridize to the opposite ends of the segment that we are interested in studying. In our case, we would add primers corresponding to a segment of our *Drosophila* gene, using the cDNA from each embryonic stage as a template for PCR amplification in separate samples. When the products are analyzed on a gel, copies of the amplified region will be observed as bands only in samples that originally contained mRNA from the gene being studied. This

method can tell researchers whether an mRNA is present. However, accurate measurement of mRNA levels requires newer, more quantitative PCR machines that use a dye that fluoresces only when bound to a double-stranded PCR product. This technique, called *quantitative RT-PCR* (*qRT-PCR*), can detect the light and measure the PCR product, thus avoiding the need for electrophoresis while providing quantitative data, a distinct advantage. Both RT-PCR and qRT-PCR can also be carried out with mRNAs collected from different tissues at one time to discover which tissue is producing a specific mRNA.

Instructors: The Scientific Skills Exercise "Analyzing Quantitative and Spatial Gene Expression Data" can be assigned in **Mastering Biology**. Students can work with data from an experiment that investigated mRNA expression using both *in situ* hybridization and quantitative RT-PCR.

Studying the Expression of Interacting Groups of Genes

A major goal of biologists is to learn how genes act together to produce and maintain a functioning organism. Now that the genomes of a number of species have been sequenced, it is possible to study the expression of large groups of genes—an approach called the *systems approach*. Researchers use what is known about the whole genome to investigate which genes are transcribed in different tissues or at different stages of development. One aim is to identify networks of gene expression across an entire genome.

To accomplish such genome-wide expression studies, today's rapid, inexpensive DNA-sequencing methods allow researchers to discover which genes are expressed by simply sequencing the cDNA samples from different tissues or different embryonic stages. This straightforward method is called **RNA sequencing**, or **RNA-seq** (pronounced "RNA-seek"), even though it is the cDNA that is actually sequenced. In RNA-seq, the mRNA (or other RNA) samples are isolated, cut into shorter, similar-sized fragments, and converted into cDNAs **(Figure 20.12)**. These short cDNA stretches are sequenced, and a computer program reassembles them, either mapping them onto the genome of the species in question (when available) or simply putting the fragments in order from scratch based on overlapping sequences of multiple RNAs.

Figure 20.12

```
    Use of RNA sequencing (RNA-seq) to analyze expression of
many genes.
```

RNA sequencing shown in five steps.

RNA-seq yields a wide range of information about expression of genes, including their level of expression.

The steps are as follows.

1.mRNAs are isolated from the tissue being studied.

2. mRNAs are cut into similar-sized, small fragments.
3. mRNAs are reverse-transcribed into cDNAs of the same size.
4. cDNAs are sequenced.
5. The short sequences are mapped by computer onto the genome sequence. The resulting data, including the number of times a sequence is present, indicate which genes are expressed in a given tissue and at what level. Short sequences include the following. G G A G A A G T C T. G A G A A G T C T G. A G T C T G C C G T. C C G T T A C T G C. C C C T G T G G G G.

Several characteristics make RNA-seq a powerful technique. First, it is not based on hybridization with a labeled probe, so it doesn't depend on knowing genomic sequences. Second, it can measure levels of expression over a very wide range. Third, a careful analysis provides a wealth of information about expression of a particular gene, such as relative levels of alternatively spliced mRNAs. In most cases, however, expression of individual genes still needs to be confirmed by RT-PCR.

An older method of genome-wide expression studies, called **DNA microarray assays**, is less powerful than RNA-seq but is still used for some applications, such as fetal testing (see Figure 14.19). A DNA microarray consists of tiny amounts of a large number of single-stranded DNA fragments representing different genes fixed to a glass slide in a tightly spaced array, or grid, of dots. (The microarray is also called a *DNA chip* by analogy to a computer chip.) Ideally, these fragments represent all the genes of an organism. The mRNAs from the cells being studied are reverse-transcribed into cDNAs (see Figure 20.10), and a fluorescent label is added so the cDNAs can be used as probes on the microarray. Different fluorescent labels are used for different cell samples so that multiple samples can be tested in the same experiment. The resulting pattern of colored dots, shown in an actual-size microarray in **Figure 20.13**, reveals the dots to which each probe was bound and thus the genes that are expressed in the cell samples being tested.

Figure 20.13

Use of microarrays to analyze expression of many genes.

Microarray assay shows mRNA from two different human tissues. Genes expressed can be in one only of either tissue, in both tissues, or in neither tissue.

Each dot in the microarray assay is a well containing identical copies of DNA fragments that carry a specific gene. The general pattern in the microarray shows many bright yellow and green dots with some red mixed in towards the top of the microarray and in another, smaller band at the bottom. There is a wide band with mostly black in the bottom third of the microarray. Genes that were expressed in tissue #1 (the red wells)

bind to red cDNAs made from mRNAs in that tissue. Genes that were expressed in tissue #2 (the great wells) bind to green cDNAs. Genes that were expressed in both tissues bind both red and green cDNAs; these wells appear yellow. Genes that were not expressed in either tissue do not bind either cDNA; these wells appear black.

Researchers extracted mRNAs from two different human tissues and synthesized two sets of cDNAs, fluorescently labeled red (tissue 1) or green (tissue 2). Labeled cDNAs were hybridized with a microarray containing 5,760 human genes (about 25% of human genes), part of which is shown in the enlargement. Red indicates that the gene in that well was expressed in tissue 1, green in tissue 2, yellow in both, and black in neither. The fluorescence intensity at each spot indicates the relative expression of the gene.

Scientists can now measure the expression of thousands of genes at one time. DNA technology makes such studies possible; with automation, they are easily performed on a large scale. By uncovering gene interactions and providing clues to gene function, DNA microarray assays and RNA-seq may contribute to a better understanding of diseases and suggest new diagnostic techniques or therapies. For instance, comparing patterns of gene expression in breast cancer tumors and noncancerous breast tissue has already resulted in more informed and effective treatment protocols (see Figure 18.27). Ultimately, information from these methods should provide a grander view of how ensembles of genes interact to form an organism and maintain its vital systems.

Determining Gene Function

Once they identify a gene of interest, how do scientists determine its function? A gene's sequence can be compared with sequences in other species. If the function of a similar gene in another species is known, one might suspect that the gene product in question performs a comparable task. Data about the location and timing of gene expression may reinforce the suggested function. To obtain stronger evidence, one approach is to disable the gene and then observe the consequences in the cell or organism.

Editing Genes and Genomes

Molecular biologists have long sought techniques for altering, or editing, the genetic material of cells or organisms in a predictable way. In one such technique, called ***in vitro*mutagenesis**, specific mutations are introduced into a cloned gene, and the mutated gene is returned to a cell in such a way that it disables ("knocks out") the normal cellular copies of the same gene. If the introduced mutations alter or destroy the function of the gene product, the phenotype of the mutant cell may help reveal the function of the missing normal protein.

In Concept 17.5, you read about the CRISPR-Cas9 system, the powerful new technique for gene editing in living cells and organisms that is taking the field of genetic engineering by storm (see Figure 17.28). This system, worked out by Jennifer Doudna **(Figure 20.14)** and Emmanuelle Charpentier, is a highly effective way for researchers to knock out a given gene in order to study what that gene does. It has already been used in many organisms, including bacteria, fish, mice, insects, human cells, and various crop plants. Modifications of the technique allow researchers to repair a gene that has a mutation. This approach is used for gene therapy, which will be discussed later in the chapter.

Figure 20.14

```
Jennifer Doudna holding a model of CRISPR–Cas9.
```

Jennifer Doudna holds a gene model.

HHMI BioInteractive Video

Jennifer Doudna: CRISPR-Cas Proteins are the Swiss Army Knife of the Genome

image

Play

image

Play

For more resources from HHMI BioInteractive, go to www.biointeractive.org.

In another application of the CRISPR-Cas9 system, scientists are attempting to address the global problem of insect-borne diseases by altering genes in insects so that, for example, they cannot transmit disease. An extra twist to this approach is to engineer the new allele so that it is much more highly favored for inheritance than the wild-type allele. This strategy is called a **gene drive** because the biased inheritance of the engineered gene during reproduction rapidly “drives” the new allele through the population.

Other Methods for Studying Gene Function

Another method for silencing expression of selected genes doesn’t alter the genome; instead, it exploits the phenomenon of ****RNA interference (RNAi)**, described in Concept 18.3. This experimental approach uses synthetic double-stranded RNA molecules matching the sequence of a particular gene to trigger breakdown of the gene’s messenger RNA or to block its translation. In organisms such as the nematode and the fruit fly, RNAi has already proved valuable for analyzing the functions of genes on a large scale. This

method is quicker than using the CRISPR-Cas9 system, but it leads to only a temporary reduction of gene expression rather than a permanent gene knockout or alteration.

In humans, ethical considerations prohibit knocking out genes to determine their functions. An alternative approach is to analyze the genomes of large numbers of people with a certain phenotypic condition or disease, such as heart disease or diabetes, to try to find differences they all share compared with people without that condition. The logic is that these differences may be associated with one or more malfunctioning genes and thus in a sense are naturally occurring gene knockouts. In these large-scale analyses, called **genome-wide association studies**, researchers look for *genetic markers*, DNA sequences that vary in the population. In a gene, such sequence variation is the basis of different alleles, as we have seen for sickle-cell disease (see Figure 17.26). And just like the coding sequences of genes, noncoding DNA at a specific locus on a chromosome may exhibit small nucleotide differences among individuals. Variations in coding or noncoding DNA sequences among a population are called polymorphisms (from the Greek for “many forms”).

Interview

Interview with Charles Rotimi: Using genomics to study health-related conditions in African-Americans

Charles Rotimi.

Among the most useful genetic markers in tracking down genes that contribute to diseases and disorders are single base-pair variations in the genomes of the human population. For more than 99% of the nucleotides in the human genome, virtually all people have the same nucleotide in each position. However, once in every 100–300 base pairs of both coding and noncoding DNA sequences are positions where the sequence varies between individuals. A single base-pair site where variation is found in at least 1% of the population is called a **single nucleotide polymorphism** (**SNP**, pronounced “snip”); a few million SNPs occur in the human genome. To find SNPs in large numbers of people, it isn’t necessary to sequence their DNA; SNPs can be detected by very sensitive microarray assays, RNA-seq, or PCR.

Once a SNP is identified that is found in all people affected by the disease being studied, researchers focus on that region and sequence it. In nearly all cases, the SNP itself does not contribute directly to the disease in question by altering the encoded protein; in fact, most SNPs are in noncoding regions. Instead, having a particular SNP associated with a disease suggests that the gene whose mutation causes the disease is located very close to that SNP on that chromosome. This closeness (*genetic linkage*; see Concept 15.3) means that crossing over between the SNP and the gene is very unlikely during gamete formation. Therefore, the SNP and gene are almost always inherited together, so the SNP acts as a genetic marker for the disease-causing

allele **(Figure 20.15)**. SNPs have been found that correlate with diabetes, heart disease, and several types of cancer, and the search is on for genes that might be involved in these and other conditions that are based on genetics.

Figure 20.15

```
    Single nucleotide polymorphisms (SNPs) as genetic markers
for disease-associated alleles.
```

The SNP linked to a normal allele is A bonded to T. The SNP linked to a disease-associated allele is C bonded to G.

This diagram depicts the same region of the genome from two groups of individuals, one group having a particular disease or condition with a genetic basis. Unaffected people have an AT pair at a given SNP locus, while affected people have a CG pair there. Once the allele is confirmed as being associated with the disease in question, the SNP that varies in this way can be used as a marker for the disease-associated allele.

The experimental approaches you have learned about thus far focus on working with molecules, mainly DNA and proteins. In a parallel line of inquiry, biologists have been developing powerful techniques for cloning whole multicellular organisms. One aim of this work is to obtain special types of cells, called stem cells, that can give rise to all types of tissues. Being able to manipulate stem cells would allow scientists to use the DNA-based methods previously discussed to alter stem cells for the treatment of diseases. Methods involving the cloning of organisms and production of stem cells are the subject of the next section.